



Article

Understanding Millennials' Financial Behavior: The Role of Fintech Adoption, Financial Literacy, and the Mediating Effect of Financial Attitudes in a Crisis-Affected Emerging Economy

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Abstract

This study investigates how financial literacy, FinTech adoption, and financial attitudes shape economic decision-making among millennials in Lebanon, a crisis-affected emerging economy. The study examines whether enhancing financial literacy can strengthen economic resilience through improved financial behavior, with financial attitudes acting as a mediator. Guided by Behavioral Finance Theory, the study employs a quantitative approach using data from 390 Lebanese millennials collected via a structured questionnaire. Structural equation modeling was applied to test direct and mediating effects. Both financial literacy and FinTech adoption were found to significantly influence millennials' financial behavior, with financial literacy emerging as the stronger predictor. The findings also revealed that financial attitude significantly mediates the link between literacy and behavior, suggesting that financial knowledge alone is insufficient without attitudinal reinforcement. This study fills a critical empirical gap in the MENA region by offering evidence from a highly under-researched, crisis-affected emerging market. It introduces an integrated model combining technological, cognitive, and attitudinal dimensions of financial behavior. The study offers practical implications for policymakers, financial institutions, and international development actors seeking to strengthen financial inclusion and household stability in similar turbulent contexts.

Keywords: FinTech; financial literacy; financial attitude; millennials; economic instability; behavioral finance; Lebanon



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1. Introduction

Financial literacy, delineated as the knowledge and skills that facilitate informed positive decision-making about budgeting, saving, investing, and managing credit (Mulla, 2022), is proposed to limit risk and exposure to debt for individuals and help them to make better investment decisions (Liu et al., 2024). Despite the ease of access to financial products, many populations have not acquired financial literacy to achieve positive financial behaviors. As such, the roles of financial literacy and financial behavior in achieving financial well-being are key to achieving positive financial outcomes (Choung et al., 2023). Fintech also contributes to improving financial behavior in budgeting, saving, and investing; however, better comprehending Fintech products leads to maximizing the use and benefits of their offerings (Alamelu, 2024). Fintech has changed the way people manage their

finances, particularly for the millennial generation, who use personal finance apps to track spending daily, create a budget, and make informed decisions that impact their financial wellness (Cardoso et al., 2024). Nguyen (2024) highlights that millennials, in developed economies, prefer Fintech services because they are easier and more efficient.

Within the Middle East and North Africa (MENA) context, notable heightened areas within the realm of behavioral finance encompass the focus on regulatory gaps, digital literacy, and trust issues (Elouaourti & Ibourek, 2024). Across the Lebanese landscape, the rise of Fintech has been hampered by the low level of financial literacy, and there is a noticeable lack of prior literature on Fintech-related usage and adoption.

Hence, this research seeks to fill this gap, and the objective is three-fold, as it aims to examine the connection between Fintech adoption and the financial behavior adoption of Lebanese millennials, along with exploring the correlation between financial literacy and the financial behavior of these targeted millennials, and to investigate whether financial attitude serves as a mediator between financial literacy and financial behavior. These variables become especially relevant in Lebanon due to the country's prolonged economic collapse, currency devaluation, capital controls, and declining trust in the financial system. Millennials increasingly rely on digital financial solutions and personal financial competencies to navigate daily financial decisions. As a result, understanding how financial literacy, financial attitudes, and FinTech adoption jointly influence financial behavior is critical in a crisis-affected context where traditional financial guidance and institutional support are limited. The context of economic turmoil, inflation, and issues in Lebanon's banking system makes it important to understand how millennials engage with digital financial services (Bejjani et al., 2024; Sawaya et al., 2023).

The three core constructs examined in this study, namely FinTech adoption, financial literacy, and financial attitude, collectively shape how millennials engage in financial behavior. Financial literacy forms the cognitive foundation enabling individuals to interpret financial information and evaluate choices. Financial attitudes capture the psychological and evaluative orientations toward money, influencing how literacy is translated into actual behavior. FinTech adoption represents the practical behavioral channel through which individuals execute, manage, and monitor financial activities. To ground this investigation theoretically, the study adopts Behavioral Finance Theory (BFT), which recognizes that individuals' financial decisions are shaped by cognitive processing, psychological biases, and contextual influences. Together, these elements align with BFT, which posits that financial behavior emerges from the interaction of knowledge, perceptions, and the tools individuals use. Accordingly, examining these constructs jointly provides a comprehensive lens for understanding millennials' financial behavior within Lebanon's crisis context.

This study makes three key contributions to the literature on financial behavior in emerging markets. First, it integrates FinTech adoption, financial literacy, and financial attitude within a unified behavioral finance model, offering a more holistic understanding of millennials' financial decision-making. Second, it provides novel empirical insights from Lebanon, a crisis-affected and institutionally fragile economy where data on FinTech use and financial behavior are scarce. Third, by validating the mediating role of financial attitudes, the study extends the BFT to account for psychological and attitudinal factors that shape financial behavior in uncertain and high-risk environments. These contributions offer both theoretical advancement and practical guidance for scholars, policymakers, and financial institutions.

Three pivotal research questions guide this study:

RQ1. What is the role of Fintech adoption in millennials' financial behavior?

RQ2. What is the role of financial literacy in millennials' financial behavior?

RQ3. Do financial attitudes mediate the relationship between financial literacy and financial behavior?

Although the conceptual relationships between financial literacy, financial attitude, and financial behavior have been widely documented, their dynamics in crisis-affected economies remain markedly underexplored. In Lebanon, a context characterized by institutional collapse, hyperinflation, and the erosion of public trust in the financial system, these behavioral relationships are likely to function differently than in stable markets. Thus, the contribution of this study is not rooted in proposing a novel theoretical path but in demonstrating how well-established behavioral mechanisms operate, and in some cases intensify, under extreme economic instability. By providing large-scale empirical evidence from a severely fragile financial environment, the study extends existing behavioral finance knowledge into settings where traditional assumptions about rationality, trust, and risk no longer hold.

2. Literature Review

2.1. Fintech Adoption

FinTech adoption is defined as the extent to which individuals accept, use, and integrate digital financial technologies, such as mobile banking, e-wallets, and online payment systems, into their financial activities (Swacha-Lech & Solarz, 2021). The swift development of Fintech has caused seismic shifts in the global financial environment, altering how consumers engage with financial services. The Fintech adoption process relies greatly on factors such as convenience, security, personalization, and cost, in addition to technology interest and social media reliance (Swacha-Lech & Solarz, 2021). Fintech has been incorporated inside banks covering digital banking, online banking, mobile banking, and neo-banks, all of which have improved financial inclusion via direct banking access, branchless banking, reducing transaction costs, and AI financial applications that provide automated budgeting and real-time financial management (Singh et al., 2024).

2.2. Financial Literacy

Financial literacy is defined as the ability to successfully manage or supervise one's finances (Huston, 2010). Lusardi and Mitchell (2014) exhibit that financial literacy gives someone the ability to make informed financial decisions, become freer from excessive debt, accumulate wealth, and plan financially. Skills like budgeting, knowing how to establish an emergency fund, knowing how to use credit inappropriately, and understanding investments and taxes are just a few examples of financial literacy skills. While financial literacy is an important skill set to acquire, many people lack the financial literacy skills to be successful and wind up with financial habits that are difficult to break (Song et al., 2023). Hence, acknowledging and improving individual financial literacy greatly contributes to reducing stress, strengthening economic stability, and leading toward success in the long run (Van Rooij et al., 2011).

2.3. Financial Behavior of Millennials

The financial behaviors of millennials are deemed as a research area of increased focus, outlining millennials' saving, spending, and investing tendencies. Previous observations indicate that millennials' financial behaviors can be affected by several aspects, with socialization agents being one aspect, along with digital financial literacy (Qamar et al., 2023; Bhatia, 2024). Furthermore, family members, educational and financial institutions, workplace, media, the situation of the country, and financial attitudes on digital financial literacy, all subsequently affect the financial decision-making skills of Gen Z and millennial generations (Qamar et al., 2023). Following Bhatia (2024), millennials' financial literacy

is coined with the use of Fintech and financial instruments, pinpointing that financial behavior is the most reliable predictor. This judgment reinforces the value of behavior versus knowledge as a driver of financial decision-making.

2.4. Financial Attitude

Financial attitude refers to an individual's psychological tendencies, beliefs, and evaluative judgments about money and financial decision-making (de Almeida et al., 2021; Baptista & Dewi, 2021). Perceptions about money are vital in determining financial behaviors and are a key factor, possibly the most important one, when studying individuals' financial behaviors, decision-making processes, and overall financial well-being (de Almeida et al., 2021). Financial attitudes are swayed by financial literacy levels, cultural influences and social influences, economic conditions, and personal experiences. Qamar et al. (2023) have shown that positive financial attitudes correlate with good behaviors, encouraging individuals to build savings and invest their money, while negative financial attitudes are associated with behaviors that trigger excessive spending, which creates financial instability. To date, financial attitudes are still especially relevant to millennials due to their ongoing experiences with Fintech and their experience during economic crises and shifts in job markets (Amnas et al., 2024). A clear understanding of financial attitudes is therefore useful in developing financial education programs and policies that improve the financial resilience and financial stability of any nation.

3. Theoretical Background: Behavioral Finance Theory

Behavioral Finance Theory (BFT) challenges the assumption of fully rational decision-making by emphasizing the role of psychological, emotional, and cognitive factors in shaping financial choices. Rather than treating individuals as purely rational optimizers, BFT suggests that their decisions are influenced by limited information processing, mental shortcuts, and prior experiences, which can lead to systematic deviations in saving, spending, borrowing, and investing behavior (Arran, 2023). In the context of this study, BFT supports the examination of financial literacy (cognitive processing), financial attitudes (psychological evaluation), and FinTech adoption (behavioral response) as interrelated determinants of millennials' financial behavior. Accordingly, BFT supports the study's hypotheses by framing financial behavior as the outcome of the interaction between knowledge (financial literacy), perceptions (financial attitude), and the adoption of financial tools (FinTech).

4. Research Context

Across the Lebanese context, Fintech's maturation period is finally establishing initiatives for digital infrastructure modernization, driven by the rapid growth of digital banking and payments. Banque du Liban's Circular 331 is intended to support Fintech innovation through incentives for equity investors in early-stage startups. Nonetheless, the foray into Fintech blockchain and cryptocurrency information is met with hesitancy due to legal uncertainty, lack of trust, and predominantly cash-based, informal economy structures. Increased internet and smartphone penetration assist in technology and Fintech adoption, as mobile wallets are proliferating in Lebanon, with OMT, Whish, and BOB Finance being widely adopted amid enduring banking restrictions (Boustani, 2020). Lending models, namely, peer-to-peer lending (P2P) lending and crowdfunding, are also emerging, although Fintech lending continues to face regulatory, as well as structural, hurdles (Tarhini et al., 2016).

The millennial generation is the most significant demographic to adopt Fintech commercial models, using mobile-based financial innovations as an alternative to traditional banking products and services within a stagnant economic environment (Abu Alrub et al.,

2020). Many Lebanese millennials demonstrate a lack of knowledge in essential areas such as budgeting, investing, and debt management. Lebanon is among the lowest in financial literacy when compared to countries like the UAE and Qatar. Financial education efforts in the region have been sporadic and generally ineffective, and Lebanon lacks consistent educational programs (Makdissi & Mekdessi, 2024). As such, this current study focuses on the financial literacy and behavior of millennials, especially concerning fintech, to help bridge gaps in decision-making in today's digital financial landscape amidst ongoing instability.

Regarding the Lebanese ecosystem, many Lebanese lost access to their savings since the crisis resulted in severe capital and banking limitations, which eroded public confidence in the financial system (Mawad et al., 2022). Following the collapse of the Lebanese pound, with a loss of over 90%, hyperinflation has reduced purchasing power considerably, with many families now living in poverty, and the informal economy has expanded tremendously. Cash transactions have become commonplace due to their anonymity and to avoid banking limits (Elhajjar, 2023). Concurrently, digital financial solutions, such as cryptocurrencies as a hedge against currency devaluation (El-Chaarani et al., 2023). Millennials have begun adjusting their financial behaviors by embracing informal, flexible, and digitally driven approaches, such as the use of mobile payments and digital wallets, despite regulatory concerns (Abu Daqar et al., 2021). While fintech does present alternative options, financial illiteracy continues to pose a risk to users by exposing them to poor financial decisions, fraud, and scams (Dermesrobian, 2023).

Bizri et al. (2018) noted that the multi-religious society of Lebanon influenced financial decisions, especially when looking at principles of Islamic finance as a proxy for financial attitudes towards banking, loans, and investment behavior among Muslim millennials. Plus, according to Abou Ltaif and Mihai-Yiannaki (2024), Lebanese millennials have become more risk-averse to various economic crises, notably the financial collapse of 2019. This risk aversion can be expressed by a preference for saving rather than spending or perhaps pursuing more moderate investment styles.

Millennials' financial mindset in Lebanon is further influenced by their socioeconomic status and education, as well as their access to financial services. As the Lebanese economy continues to face enormous hardship (e.g., hyperinflation, a depreciated currency, and political disequilibrium), Lebanese millennials' financial orientation is majorly affected. Many millennials lack access to traditional financial products or credit and are exploring alternatives outside the conventional financial sector to save and invest or access loans (Mawad et al., 2022; Domat, 2022). For example, Lebanese millennials are likely to engage in modern finance, such as budgeting, investing, and retirement planning, yet because many are unemployed or underemployed, they do not have the resources to save or invest, which creates disconnection and division in financial orientation and perspective (El-Chaarani et al., 2023).

In a nutshell, although Fintech adoption is growing, challenges like frugality, limited financial education, and unclear regulations continue to affect Lebanese millennials' financial behavior. The ongoing economic challenges only make these issues more difficult. Hence, there is a clear need for better financial education and clearer regulations to help millennials take full advantage of entrepreneurial opportunities. All these conditions make Lebanon an appropriate setting for analyzing how knowledge, attitudes, and digital financial tools collectively shape financial behavior.

5. Hypotheses Development

5.1. Fintech Adoption and Financial Behavior

Prior research posits that financial behavior is closely linked to Fintech adoption. For instance, Sreelakshmi (2024) argues that the use of Fintech significantly impacts financial

behavior. Further, [Aftab et al. \(2025\)](#) evidenced that risk-averse and financially overconfident consumers are less likely to adopt Fintech, outlining some of the customers' financial tendencies and trends associated with Fintech usage. In this backdrop, [Yang and Zhang \(2022\)](#) consolidate that higher FinTech adoption amplifies consumption across households. In turn, [Gafoor and Amilan \(2024\)](#) establish that Fintech adoption sways individuals' financial well-being and behavior. Further, [Yousef \(2024\)](#) found that an increase in e-wallet use resulted in an upsurge in both purchasing behavior and financial management. Likewise, [Abu Daqar et al. \(2021\)](#) explored fintech adoption among the Millennial and Gen Z generations, postulating that millennials display a high tendency to use e-wallet services to manage their business payroll and pay their tuition fees via their e-banking or Fintech services. These studies summarize key factors that influence fintech adoption and highlight its role in promoting trends associated with financial behavior through these digital financial services. From a behavioral finance perspective, FinTech adoption represents a behavioral tool through which individuals act on their financial decisions, supporting the expectation of a positive relationship with financial behavior. Hence, the following hypothesis is presented as follows:

H₁. *Fintech adoption positively impacts the financial behavior of millennials.*

5.2. Financial Literacy and Financial Behavior

Various studies have examined the link between financial literacy and financial behavior. For instance, [Rahayu et al. \(2023\)](#) found that financial literacy positively impacts financial management behavior, and [Aftab et al. \(2025\)](#) elucidate the positive role of financial literacy in encouraging consumers to incorporate Fintech services as part of their financial behavior. Further, [García Mata \(2021\)](#) and [Singh et al. \(2024\)](#) intend that financial knowledge is significantly associated with financial well-being and personal finance management. Similarly, [Maalouf et al. \(2023\)](#) analyzed Lebanese university students' financial tendencies and found that financial literacy increases sensible financial decision-making and behavior. [Khan et al. \(2024\)](#) found that financial literacy strongly influences consumer behavior, especially concerning debt, and highlighted financial education efforts aimed at improving not only financial well-being but mental well-being as well. Based on previous insights, and that BFT posits that higher financial literacy reduces cognitive constraints and improves judgment, supporting its expected positive effect on financial behavior, a second hypothesis is added to the model as follows:

H₂. *Financial literacy positively impacts the financial behavior of millennials.*

5.3. Financial Attitude, Financial Literacy, and Financial Behavior

Studies have focused on the interchange between financial attitude, financial literacy, and financial behavior. For instance, [Baptista and Dewi \(2021\)](#) concluded that both financial literacy and attitudes exert a significant influence on financial management behavior. In parallel, [Allen \(2023\)](#) found that financial literacy and adopting FinTech positively affect financial behavior while emphasizing that financial attitude is a component in making a financial decision. Furthermore, previous observations entail that both financial literacy and financial attitude affect behavioral finance and consumptive behavior, with financial attitude mediating the association between financial literacy and financial behavior ([Nano, 2015](#); [Muslikhun & Wahjoedi, 2023](#); [Widjayanti et al., 2025](#); [Wahyun et al., 2023](#)). As such, in line with previous inquiries and drawing on BFT, financial attitude is understood as the psychological mechanism through which knowledge is translated into real financial choices, justifying its mediating role. The following hypotheses are anticipated:

H₃. *Financial literacy positively impacts the financial attitude of millennials.*

H₄. Financial attitude positively impacts the financial behavior of millennials.

H₅. Financial attitude mediates the relationship between financial literacy and the financial behavior of millennials.

Accordingly, Figure 1 illustrates the study's conceptual framework, outlining the assumed associations among the studied variables as well as the anticipated hypotheses.

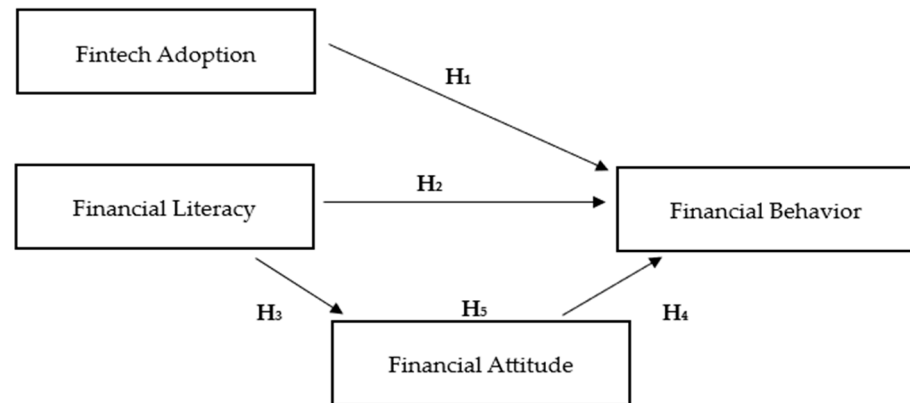


Figure 1. Research Model. Source: Authors.

6. Methodology

6.1. Research Design

A positivist philosophy was followed, and the reasoning in this explanatory study is deductive, alongside a mono-quantitative methodological choice. As such, a self-report survey was employed using closed-ended questions for demographics, and five-point Likert scale statements for the studied variables (1 = Strongly disagree, 5 = Strongly agree). Given the absence of publicly available national data on Fintech adoption, financial literacy, financial attitudes, and financial behavior in Lebanon, the use of a primary questionnaire was both necessary and justified. Moreover, considering the relevance and urgency of the topic amid Lebanon's ongoing economic crisis, this approach provides timely, context-specific insights into a critically underexplored population segment.

The survey was split into five parts. The first part covers demographic information encompassing background insights related to their age, gender, educational background, employment status, and income. The second part reflects the reported usage and adoption of Fintech (FITAD). The third part is related to the participants' financial literacy (FINL) measurements. The questionnaire's fourth part is related to the financial attitude of Lebanese millennials (FINA). Lastly, the fifth part is reserved for respondents' financial behavioral conduct (FINB), following the behavioral finance theory. Each scale was preferred to provide a robust understanding of the associations between FITAD, FINL, FINA, and FINB in the context of this study.

Therefore, FITAD is examined according to five statements, retrieved from [Ahmad et al. \(2021\)](#) observations on the frequency or confidence of using Fintech services, such as online banking, mobile payments, and digital wallets. In turn, FINL items are adapted from [Potrich et al. \(2025\)](#), translating the understanding of financial concepts like interest rates and inflation planning for the future with finances, and covering four statements. FINA items are studied based on the study of [Peach et al. \(2017\)](#) on attitudes towards money, like money saving, spending, or taking financial risks, comfort with using debt, and encompass four statements. Lastly, FINB items are adapted from [OECD \(2022\)](#), which collected participants' stratified responses about financial behavior, budgeting, saving, repayment of debt, and investing habits.

Before data collection, a pilot test was performed on 20 professionals to secure clarity and reliability. Subsequently, survey participants were provided with substantial information related to this research aim, data utilization, and their rights as research subjects. Respondents then validated their written informed voluntary consent for participating in the questionnaire, and ethical consideration was ensured upon server confidentiality. Also, the names of the survey participants were not supplied to avoid privacy issues. Lastly, ethical approval was obtained from the Research Ethics Committee (REC) at the Arab Open University (AOU)-Lebanon under the reference AOU-IRB-2025-100.

6.2. Population and Sample

By 2025, the estimate indicates that the population in Lebanon will be 5.81 million, relatively higher than that of 2024 due to stability and prosperity within the country ([Worldometer, 2024](#)). Millennials, born between 1981 and 1996, and aged between 28 and 43, make up around 23.6% of the Lebanese total population ([DataReportal, 2024](#)). This study utilizes random sampling to survey millennials and follows up with snowball sampling. According to Qualtrics, the reliable sample size used for this research is 385 respondents. Hence, online survey data were collected from Lebanese millennials by distributing the questionnaires using Google Forms from March 2025 to May 2025. A final data set of 390 participants was obtained, and the distilled data were then analyzed using JASP statistical software version 0.95.4.0 to test the hypothesized relationships among the variables.

6.3. Data Analysis

In this investigation, Structural Equation Modeling (SEM) was employed to validate the presumed hypotheses showcased in the conceptual model. SEM was selected because it allows simultaneous estimation of multiple relationships among latent constructs, accounts for measurement error, and is suitable for testing mediating effects. Given the multidimensional nature of behavioral finance constructs, SEM provides stronger explanatory power compared to traditional regression methods. A Confirmatory Factor Analysis (CFA) was conducted before the structural model to confirm that all items loaded significantly onto their respective constructs and that the measurement model met the required validity and reliability thresholds.

Model robustness and validity were assessed following standard SEM procedures. Internal consistency was examined using Cronbach's alpha and composite reliability, while convergent validity was evaluated through average variance extracted (AVE). Discriminant validity was confirmed using the Heterotrait–Monotrait (HTMT) ratio.

Although socio-demographic variables were collected to describe the sample, they were not included in the structural model. The study followed a theoretically driven modeling strategy based on BFT, which focuses primarily on cognitive, attitudinal, and behavioral constructs. Including demographics, without theoretical grounds linking them to the endogenous variables, would add unnecessary complexity and reduce model parsimony. For this reason, socio-demographic variables were retained for descriptive purposes only and not incorporated into the final SEM model.

7. Results

7.1. Sample Profile

Table 1 reveals the socio-demographic insights of the obtained sample size. Socio-demographic variables were collected for descriptive purposes but were not included in the SEM analysis, consistent with the study's theory-driven model specification.

Table 1. Sample Profile.

Category	Subcategory	Percentage (%)
Gender	Female	47.2
	Male	43.1
	Prefer not to say	9.7
Age	between 18 and 25 years	11.3
	between 26 and 35 years	42.6
	between 36 and 45 years	24.1
	between 46 and 55 years	13.1
	56 and Above	8.9
Educational Background	High School	13.8
	BA Degree	40.5
	Master's Degree	35.2
	Doctorate Degree	10.5
Marital Status	Single	49.2
	Married	33.3
	Divorce	13.6
	Widowed	3.9
Employment Status	Employed	60.5
	Self-Employed	22.1
	Unemployed	14.1
	Retired	3.3
Income	Below \$500	21.3
	Between \$501 and \$1000	29.2
	Between \$1001 and \$1500	13.6
	Between \$1501 and \$2000	12.1
	Higher than \$2000	11.5
	Prefer not to disclose	12.3

7.2. Factor Analysis

Table 2 depicts an overall Kaiser–Meyer–Olkin (KMO) value of 0.963, pointing out the high appropriateness of the collected data for factor analysis. Plus, Bartlett's test shows a chi-square (χ^2) of 5218.545 and a p -value lower than 0.001, implying that there exist correlations between the measured variables, which proves the vitality of proceeding with factor analysis.

Table 2. KMO Test.

Indicator	MSA
FINL1	0.965
FINL2	0.962
FINL3	0.957
FINL4	0.956
FINA1	0.957
FINA2	0.953
FINA3	0.964
FINA4	0.964
FITAD1	0.961
FITAD2	0.977
FITAD3	0.967
FITAD4	0.962
FITAD5	0.918
FINB1	0.978

Table 2. *Cont.*

Indicator		MSA
FINB2		0.973
FINB3		0.975
FINB4		0.979
FINB5		0.972
Overall		0.963
Bartlett's test of sphericity		
X ²	df	p
5218.545	153	<0.001

7.3. Validity and Reliability

The AVE values generally outline the amount of variance a measured variable gets from its indicators relative to measurement error. An AVE value higher than 0.50 indicates that a construct describes more than 50% of the variance of its items, underscoring good convergent validity. In this study, all four factors have AVE values significantly greater than 0.50 (Table 3), indicating good convergent validity.

Table 3. AVE.

Factor	AVE
Factor 1	0.664
Factor 2	0.662
Factor 3	0.718
Factor 4	0.563

Also, the HTMT ratio, which determines the discriminant validity, ensuring that the measured constructs are distinct from each other, as illustrated in Table 4, most HTMT values are lower than 0.85, positing significant discriminant validity among variables.

Table 4. HTMT Ratio.

Factor 1	Factor 2	Factor 3	Factor 4
1.000			
0.682	1.000		
0.730	0.735	1.000	
0.840	0.873	0.803	1.000

Similarly, all values of coefficient α and coefficient ω are greater than 0.7 showcased in Table 5, implying that the developed constructs are significantly reliable.

Table 5. Reliability.

	Coefficient ω	Coefficient α
Factor 1	0.888	0.887
Factor 2	0.887	0.886
Factor 3	0.931	0.925
Factor 4	0.865	0.866
total	0.959	0.954

7.4. Assessment of the Measurement Model

Table 6 shows the model fit indices, which indicate that the model is a good fit.

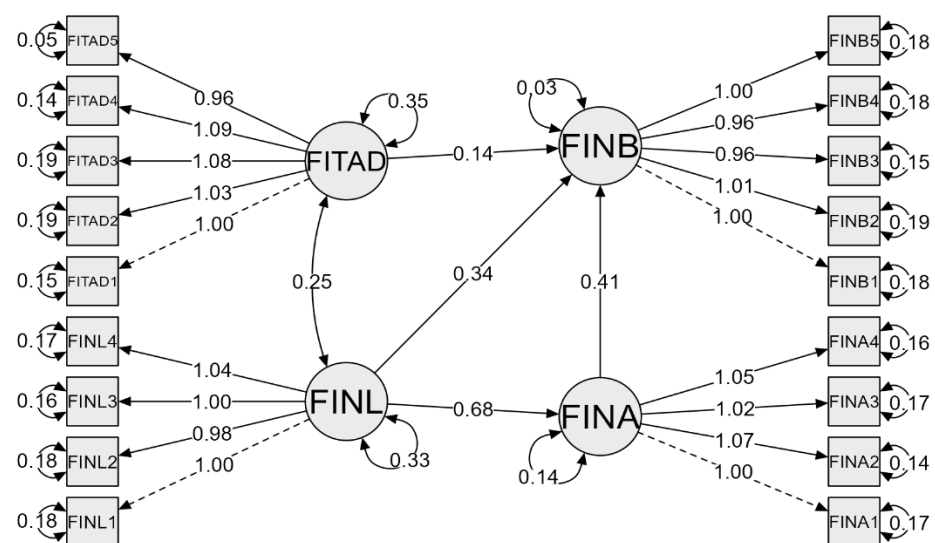
Table 6. Model Fit.

Index	Threshold	Value Obtained	Interpretation
CFI (Comparative Fit Index)	≥ 0.90	0.908	Good fit.
TLI (Tucker–Lewis Index)	> 0.90	0.921	Good fit.
RFI (Relative Fit Index)	> 0.90	0.915	Good fit.
RNI (Relative Noncentrality Index)	> 0.90	0.918	Good fit.
SRMR (Standardized Root Mean Square Residual)	< 0.08	0.022	Good fit.
RMSEA (Root Mean Square Error of Approximation)	< 0.05	0.040	Good fit.

The obtained SEM regression results are displayed in Table 7, as well as the path diagram illustrated in Figure 2.

Table 7. SEM Results.

Predictor	Outcome	Estimate	Std. Error	z-Value	p
FINL	FINA	0.681	0.053	12.737	< 0.001
	FINB	0.337	0.059	5.729	< 0.001
FINA	FINB	0.414	0.05	8.366	< 0.001
FITAD	FINB	0.144	0.041	3.48	< 0.001

**Figure 2.** Path Diagram.

First, millennials' fintech adoption positively predicts their financial behavior ($\beta = 0.144, p < 0.001$). This result confirms that the usage and adoption of financial technologies like e-wallets or online banking improves millennials' financial management and strongly defines their financial behavioral conduct. Hence, the more they engage with fintech-like financial services, the better they can manage their finances.

Second, millennials' financial literacy also positively influences their behavior ($\beta = 0.337, p < 0.001$). The following statistical result suggests that higher financial literacy leads to better financial decisions. Thus, the more millennials are knowledgeable about financial services and technological advancement, as well as financial risks, the more their financial behavior, including decision-making, is positively swayed.

Thirdly, the financial attitude of millennials is found to be a strong and positive predictor of their behavior ($\beta = 0.414, p < 0.001$), implying that attitude towards financial

services highly determines behavioral conduct. In turn, FINL significantly impacts FINA ($\beta = 0.681, p < 0.001$), postulating that financial attitude partially mediates the correlation between financial literacy and financial behavior and indicating that FINL alone weakly predicts millennials' FINB, yet when combined with FINA, it better shapes the financial running and decisions of millennials.

Thus, FINA is positioned as a mediator of the established association between FINL and FINB of millennials and the strongest predictor of FINB ($\beta = 0.414, p < 0.001$), in contrast to FITAD and FINL's relatively lower impacts on FINB. These results imply that millennials' financial tendencies and activities are mainly driven by their attitude towards financial services and technologies, alongside their financial knowledge, as well as usage and adoption of Fintech, which contribute to managing their finances as well.

8. Discussion

8.1. Association Between Fintech Adoption and Financial Behavior

Millennials' Fintech adoption is positively and directly correlated with their financial behavior ($\beta = 0.144, p < 0.001$), supporting H_1 and anticipating that the use and adoption of Fintech services enhance millennials' financial management and organize their financial tendencies. This aligns with prior observations depicted in the studies of [Abu Daqar et al. \(2021\)](#), [Swacha-Lech and Solarz \(2021\)](#), [Yang and Zhang \(2022\)](#), [Sreelakshmi \(2024\)](#), [Yousef \(2024\)](#), [Gafoor and Amilan \(2024\)](#), and [Aftab et al. \(2025\)](#), outlining that the more millennials engage with financial technologies, the more beneficial it is for them to manage their finances and the more their financial behavior is enhanced. As highlighted by [Swacha-Lech and Solarz \(2021\)](#) and [Abu Daqar et al. \(2021\)](#), both trust and ease of use as common motivational drivers of Fintech adoption.

8.2. Association Between Financial Literacy and Financial Behavior

Millennials' financial literacy is positively and significantly associated with their financial behavior ($\beta = 0.337, p < 0.001$), supporting H_2 and implying that higher levels of financial literacy contribute to making better financial decisions among millennials. This statistical result is congruous with previous scholars' conclusions, delineating that the more awareness spread among millennials regarding financial services and technological advancements, along with financial risks, the better the financial behavioral conduct and decision-making are shaped ([García Mata, 2021](#); [Rahayu et al., 2023](#); [Maalouf et al., 2023](#); [Singh et al., 2024](#); [Khan et al., 2024](#); [Aftab et al., 2025](#)). Notably, the captured finding is consistent with the study of [Maalouf et al. \(2023\)](#) conducted in the Lebanese context, as the authors noted that Lebanese students with higher financial literacy make better financial decisions.

8.3. Association Between Financial Attitude, Financial Literacy, and Financial Behavior

Financial literacy significantly impacts millennials' financial attitude ($\beta = 0.681, p < 0.001$), accepting H_3 , and in turn, financial attitude strongly and positively predicts millennials' financial behavior ($\beta = 0.414, p < 0.001$), supporting H_4 and implying that financial attitude is a strong mediating variable to the relationship between financial literacy and financial behavior (H_5). This observation is in harmony with previous results demonstrating that both financial literacy and financial attitude positively influence behavioral finance and consumptive behavior and that financial attitude mediates the causal linkage between financial literacy towards financial behavior ([Nano, 2015](#); [Baptista & Dewi, 2021](#); [Allen, 2023](#); [Muslikhun & Wahjoedi, 2023](#); [Wahyun et al., 2023](#); [Widjayanti et al., 2025](#)). Thus, this finding emphasizes the mediating role of mindset and attitude in shaping the financial behavior of millennials.

While many of these findings are consistent with prior research conducted in financially stable environments, notable differences emerge in the Lebanese context. For example, the unusually strong mediating effect of financial attitude contrasts with evidence from developed markets, where financial literacy often more directly predicts behavior (e.g., García Mata, 2021; Singh et al., 2024). In Lebanon, the attitudinal component exerts an amplified influence, likely because decision-making takes place under extreme uncertainty, diminished trust, and heightened perceived vulnerability. Similarly, although FinTech adoption positively influences financial behavior, its effect size is weaker than what has been observed in digitally advanced economies, suggesting that infrastructural fragility, regulatory ambiguity, and mixed trust in digital platforms limit the behavioral potential of FinTech. These contrasts highlight that well-established relationships in international literature operate differently in crisis-affected settings.

Cultural dynamics further shape these behavioral patterns. Lebanese society remains heavily influenced by cash dependency, strong intergenerational financial support systems, and religious norms that guide saving, borrowing, and investment practices. These cultural anchors often serve as substitutes for weakened financial institutions, especially after the 2019 banking crisis. As a result, millennials may rely more on subjective judgments, family influence, and informal financial networks than on formal financial knowledge, thereby strengthening the role of financial attitudes relative to objective financial literacy. These cultural considerations provide additional explanatory depth and reinforce why Lebanese financial behavior diverges from patterns commonly recorded in Western or economically stable contexts.

9. Conclusions

This study examined how FinTech adoption, financial literacy, and financial attitudes jointly shape the financial behavior of Lebanese millennials within a prolonged economic and institutional crisis. The findings confirm that financial literacy significantly enhances financial behavior, both directly and indirectly, through a strong mediating effect of financial attitude. FinTech adoption also contributes positively, though with a smaller effect size, reflecting the infrastructural, regulatory, and trust limitations characteristic of Lebanon's financial environment. While several findings align with global behavioral finance patterns, others are distinctly shaped by Lebanon's prolonged crisis and institutional breakdown. The analysis, therefore, distinguishes between mechanisms that are generalizable and those that are specific to the Lebanese context. The results demonstrate that in contexts marked by instability, currency devaluation, and weakened banking systems, financial behavior becomes more strongly influenced by psychological evaluations and attitudinal mechanisms than in stable economies. This provides important empirical evidence on how behavioral finance relationships operate differently under crisis conditions.

9.1. Theoretical Implications

In crisis-affected environments, the traditional pathways proposed by BFT undergo significant contextual modification. Crisis-induced uncertainty, survival-driven heuristics, and the erosion of institutional trust reshape the magnitude and direction of established behavioral relationships. Under such conditions, financial attitudes acquire disproportionate influence because individuals rely more heavily on subjective judgments, emotional coping strategies, and heuristic shortcuts when formal financial systems are unstable. These mechanisms help explain why the literacy–attitude–behavior pathway is amplified in Lebanon. Thus, the study extends the BFT by illustrating how macroeconomic stress, institutional breakdown, and high-risk environments distort otherwise stable behavioral mechanisms. In addition, the findings extend BFT by illustrating how cognitive, psychological, and

contextual pressures interact in crisis economies, offering a theoretical contribution relevant for research in other fragile markets.

9.2. Implications for Practitioners

This research offers direction for both action and implications for policymakers and educational and financial institutions.

For policymakers, the findings underscore the urgent need to rebuild financial trust through transparent regulation, credible communication, and protection against financial misconduct, elements that strongly shape attitudes in crisis environments. For financial institutions and FinTech providers, the results suggest that digital financial engagement must be supported by intuitive, low-risk tools that address users' heightened sensitivity to uncertainty. Integrating financial education within mobile apps, reducing friction in onboarding processes, and ensuring clear data protection mechanisms are essential to influencing attitudes positively. For international development organizations, the evidence indicates that financial literacy programming in fragile states must prioritize emotional and attitudinal components, not only technical knowledge, to support households navigating prolonged instability. These insights extend beyond Lebanon and apply broadly to fragile economies experiencing institutional breakdown, inflationary shocks, or trust erosion.

Further, areas of the regulatory framework need to be reinforced to build trust in Fintech platforms, advocating for their adoption, especially in the areas of cybersecurity, data protection, and digital transactions. Financial institutions can also impact the building of positive financial attitudes and behaviors through enhanced affordability and innovation. These institutions are urged to develop programming that is user-friendly and utilizes a Fintech lens, with educational components from budgeting tools with tutorials to personalized financial advice. For engagement to occur, market incentives in the form of reduced transaction fees, loyalty rewards, or benefits that are tiered based on the usage of digital services can lower the barrier to adopting Fintech approaches. In addition, educational institutions are prompted to develop more foundational and applied financial literacy curricula and develop programs centered around important life skills, such as budgeting skills, saving, investing, and borrowing. As a result, the confidence and proficiency of those taught in managing a new world of digital experiences are enhanced.

In turbulent environments and crisis-affected economies, empowering millennials with both tools, Fintech, and mindset (attitudes) is vital to reinforce financial resilience. Community-based peer learning and success stories should be encouraged to normalize responsible financial behavior among young adults.

9.3. Limitations and Future Research

Despite the valuable insights this study delivers, it has a few limitations worth mentioning.

Initially, the study's sample limits the generalizability of the study, considering that millennials in Lebanon grew up in an exceedingly heterogeneous population. Also, the cross-sectional design limits tracking changes in financial behavior, so further interventions can conduct longitudinal studies that would have significant improvements. The reliance on self-reported surveys also presents a limitation, as self-reports of financial behavior may reflect what participants perceive as their tendencies, yet are not indicators of genuine behavior. Although the study employed a quantitative approach, incorporating qualitative methods such as interviews could yield rich insights that better contextualize user experiences and motivations related to Fintech adoption.

A further limitation concerns the measurement approach used for financial literacy and financial behavior. Both were assessed exclusively through self-reported Likert-scale items, which reflect perceived tendencies and subjective confidence rather than objectively verified

financial knowledge or actual behavioral practices. This reliance on perception-based indicators may inflate or distort associations among variables due to self-assessment bias or social desirability effects. Future research should therefore integrate objective measures of financial literacy, such as interest-rate calculations, inflation comprehension, numeracy tasks, and risk-diversification concepts, as well as behavior-based indicators including saving regularity, budgeting habits, payment-method usage, and digital transaction records. Combining subjective and objective measures would substantially strengthen construct validity and provide a more accurate assessment of financial capabilities and real-life financial behavior.

Furthermore, although socio-demographic variables were collected, they were not incorporated into the structural model as control variables. Future studies may examine whether demographic differences moderate the relationships among the constructs.

Additionally, although this study is positioned within Lebanon's severe economic crisis, it does not incorporate key crisis-specific constructs such as institutional trust, perceived financial risk, inflation exposure, or perceived banking instability. These variables are central to understanding financial behavior in fragile economies, yet they were not included due to the study's focus on the behavioral pathways proposed by BFT and the lack of validated crisis-specific scales at the time of data collection. Their omission limits the ability to draw direct causal inferences about how macro-level instability shapes micro-level financial decisions. Future research should therefore integrate these constructs to more fully capture the psychological and structural mechanisms operating in crisis environments and to enhance the explanatory power of behavioral models in high-uncertainty settings.

Future research could strengthen and extend these findings through cross-country comparative analyses, particularly by contrasting Lebanon with more financially stable economies (e.g., UAE or Qatar) or with other crisis-affected markets. Such comparisons would clarify whether the behavioral mechanisms identified in this study are context-specific or generalizable. Further studies could also explore generational differences (e.g., Millennials vs. Gen Z).

Taken together, these methodological and conceptual constraints highlight the need to interpret the findings with caution. While the study provides meaningful insights into behavioral mechanisms among millennials in a crisis-affected economy, the scope of the analysis remains limited by the selected constructs, the reliance on self-reported data, and the absence of contextual crisis-related variables. As such, the study offers a focused but not exhaustive account of financial behavior in Lebanon. Future research should adopt more comprehensive designs, integrating objective measures, longitudinal data, and additional crisis-specific constructs, to develop a fuller and more robust understanding of financial decision-making in fragile economic environments.

9.4. Contributions to the Broader Literature

This study advances the literature on financial behavior in emerging markets in four distinct ways, offering primarily methodological and empirical contributions that extend beyond the Lebanese context.

First, it goes beyond typical unidimensional approaches in financial behavior research by demonstrating that cognitive, psychological, and technological factors must be examined jointly rather than in isolation. While existing studies typically focus on financial literacy (cognitive), attitudes (psychological), or FinTech adoption (technological) as standalone predictors, this research provides an integrated framework that captures their simultaneous and interactive effects. This multidimensional perspective offers a more comprehensive and realistic representation of how financial decisions are formed, particularly in contexts where traditional financial guidance systems are absent or inaccessible.

Second, the study contributes methodologically by providing a validated and replicable structural model specifically calibrated for crisis-affected markets. Most behavioral finance models are developed and tested in stable economic environments, limiting their applicability to fragile or volatile contexts. By demonstrating that established measures and relationships remain empirically robust even under severe macroeconomic stress, although with altered magnitudes, this research offers a methodological template that can be adapted and applied to other emerging economies experiencing institutional breakdown, inflationary shocks, or financial system collapse. The structural equation modeling approach used here provides a replicable analytical framework for researchers seeking to investigate financial behavior in similar high-risk environments.

Third, this research addresses a persistent gap in global financial behavior studies, which have historically concentrated on developed economies or stable emerging markets. By supplying large-scale primary data from a severely underrepresented region, the Arab world during an acute crisis, the study expands the empirical foundation upon which global behavioral finance theories are built. This contribution is particularly significant given the near-total absence of nationally representative statistics on FinTech usage, financial literacy levels, or behavioral outcomes in Lebanon and similar MENA countries. The evidence generated here enables more inclusive and geographically diverse meta-analyses and comparative studies, reducing the Western-centric bias that has long characterized the field.

Fourth, the findings hold transferable value for scholars, policymakers, and practitioners working in parallel contexts marked by economic volatility, rapid digital transformation, and generational shifts in financial engagement. The mechanisms identified here, particularly the amplified role of attitudes under uncertainty and the conditional effectiveness of digital financial tools, offer empirical reference points for cross-country comparative research. Countries experiencing hyperinflation, banking sector instability, or erosion of institutional trust, such as Argentina, Venezuela, Zimbabwe, or Sri Lanka, can draw on these findings to inform both academic inquiry and policy design. This cross-contextual applicability positions the study as a useful reference for the growing body of research on financial resilience and digital financial inclusion in fragile states.

Collectively, these contributions move the field forward by integrating dimensions that are often studied separately, by providing methodological scaffolding for crisis-economy research, by filling critical empirical voids in underrepresented regions, and by offering insights that resonate across diverse yet structurally similar economic contexts.

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